

Aryan Bhagat

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SUMMARY

Software Engineer with experience building backend systems, distributed event-driven applications, APIs, and production-oriented AI workflows. Skilled in Python, Java, C++, Kafka, Spring Boot, FastAPI, Docker, REST APIs, LangChain, and cloud deployment. Strong foundation in data structures, system design, debugging, and scalable software engineering.

EDUCATION

California State University, East Bay | *M.S. in Computer Science*

Aug 2024 – May 2026

Gujarat Technological University | *B.E. in Computer & Electronics Engineering*

Oct 2020 – May 2024

WORK EXPERIENCE

AI Engineer Intern | *Simple Ticket* | Boston, MA

Jul 2025 – Sep 2025

- Developed **AI CRM chatbots** with OCR-based document ingestion, semantic retrieval, and **RAG pipelines** using **LangChain**, **LangGraph**, and vector databases.
- Built API-driven agent workflows using **REST APIs**, **LangChain Expression Language**, and tool orchestration to support stateful multi-step AI execution.
- Containerized AI services using **Docker** and integrated asynchronous API workflows for scalable deployment and low-latency inference pipelines.

Research Assistant | *California State University, East Bay* | Hayward, CA

Sep 2025 – Mar 2026

- Developed robotic perception pipelines on **JetAuto Pro** and **Jetson Orin Nano** using **ROS2**, **OpenCV**, and 3D depth cameras for real-time perception, object tracking, and autonomous navigation workflows.
- Integrated **LiDAR spatial mapping**, AR visualization, and sensor fusion techniques to improve navigation accuracy and reduce end-to-end system latency by **15%** in dynamic environments.
- Optimized real-time sensor processing pipelines using **Python**, **ROS2**, and multithreaded data handling techniques to improve perception throughput, reduce inference bottlenecks, and stabilize concurrent sensor communication.

Firmware Engineer | *Indieseem Private Limited* | Ahmedabad, India

Jan 2024 – Jun 2024

- Developed embedded applications on **Nordic nRF52/nRF53 DKs** and **Quectel EVKs** using **Zephyr RTOS**, Nordic nRF Connect SDK, Segger J-Link, and RTT debugging.
- Architected a **BLE communication framework** with UART, SPI, I2C, and LoRa support, enabling concurrent peripheral communication and improving system throughput by **10%**.

PROJECTS

Distributed Event-Driven Notification Platform

[GitHub](#)

- Architected a Kafka-based distributed notification platform using **Apache Kafka**, **Spring Boot**, and microservices with priority-isolated topics, consumer groups, and concurrent delivery pipelines.
- Implemented asynchronous event pipelines with retry handling, priority-based routing, and external API orchestration for fault-tolerant Email, SMS, and Push notification delivery.

Smart Recipe Tagger – Multimodal AI Recipe Platform

[GitHub](#)

- Built a full-stack platform using **React**, **Node.js**, **MongoDB**, **Firebase**, and Google OAuth for image ingestion, recipe generation, and user content management.
- Designed a multimodal inference pipeline using **Google Vision API** and **Gemini API**, deployed with **Docker** and **CI/CD workflows** on Google Cloud Run.

FactShield – Ensemble Fake News Detection Platform

[GitHub](#)

- Built a Flask-based inference service integrating **RoBERTa**, **XGBoost**, **LightGBM**, and **Scikit-learn** for real-time misinformation classification.
- Designed a weighted multi-model inference engine combining transformer-based semantic analysis with traditional ML classifiers for confidence-scored predictions.

RAG-Based Personalized Outreach Engine

[GitHub](#)

- Developed a retrieval-augmented outreach platform using **Llama 3.3**, **Groq**, **LangChain**, **ChromaDB**, and **Streamlit** to generate context-aware cold emails.
- Built pipelines for web scraping, semantic embeddings, and vector similarity search to retrieve company and role-specific context for grounded LLM outputs.

SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, Go

Backend & Distributed Systems: Spring Boot, FastAPI, Flask, Node.js, Apache Kafka, REST APIs, Microservices, Event-Driven Architecture, WebSockets

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, RAG, OpenCV, YOLO, MLflow

Databases & Cloud: PostgreSQL, MongoDB, ChromaDB, AWS (EC2, S3), Google Cloud Run, Docker, CI/CD, Git, GitHub

Engineering: Data Structures, Algorithms, OOP, System Design, Distributed Systems, Debugging, Profiling, Testing